Unit 8

Orthogonality and projection.

Unit Goals

- Be able determine if two vector spaces are orthogonal.
- Be able to explain what is meant by the projection of a vector onto a subspace.
- Be able compute the projection of a vector onto a subspace.
- Be able to use the projection to determine the error in an approximation solution to a linear system.
- Be able to find the best approximate solution to an inconsistent linear system.

Unit 8.1

Orthogonal vector spaces

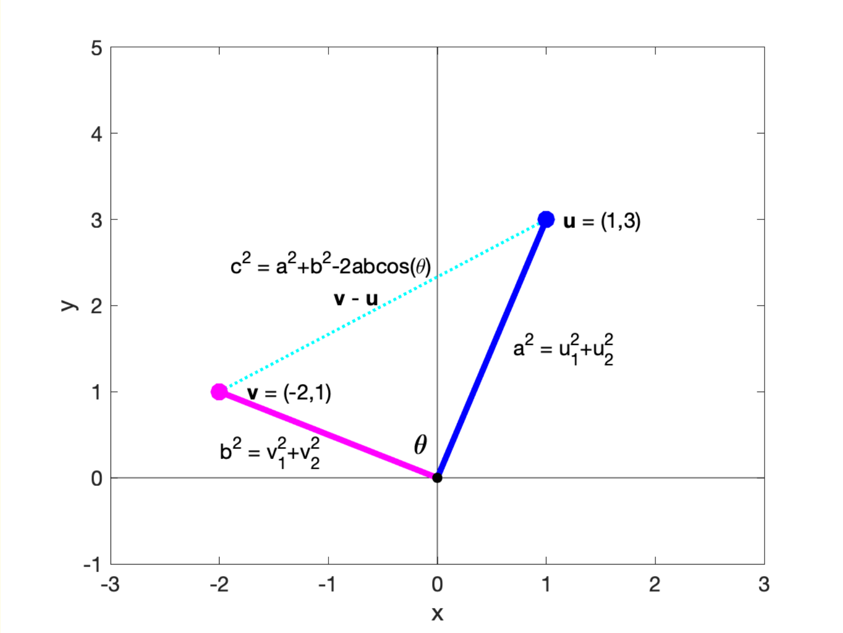
What is orthogonality?

Definition:

- Two vectors $\vec{u}$ and $\vec{v}$ are **orthogonal** to one another if and only if their dot product is zero:

$$\vec{u} \cdot \vec{v} = 0 \leftrightarrow \vec{u}^T \vec{v} = 0$$

What is orthogonality?



- **The Cosine Law:**

$$\|\vec{v}-\vec{u}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\|\vec{u}\|\|\vec{v}\|\cos\theta$$

$$\cos\theta = \frac{\|\vec{u}\|^2+\|\vec{v}\|^2-\|\vec{v}-\vec{u}\|^2}{2\|\vec{u}\|\|\vec{v}\|}$$

$$\cos\theta = \frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\|\vec{v}\|}$$

- **Two non-zero vectors are orthogonal to one another if and only if they lie at right angles (π/2 rad or 90°) to each other.**

$$\vec{u}\cdot\vec{v} = 0 \quad \rightarrow \quad \cos\theta = 0 \quad \rightarrow \quad \theta = \frac{\pi}{2}$$

- **The zero vector is orthogonal to every vector, including itself:**

$$\vec{u}\cdot\vec{0} = 0, \qquad \forall\vec{u}$$

What is orthogonality?

Definition:

- Two vector spaces U and V are **orthogonal** to one another if and only if the dot product of every vector from U with every vector from V is zero:

$$U \perp V \leftrightarrow \vec{u}^T \vec{v} = 0, \qquad \forall \vec{u} \in U, \qquad \forall \vec{v} \in V$$

Notation: "$\perp$" is read as "is orthogonal to"

What is orthogonality?

- Two vector spaces U and V are orthogonal to one another if and only if every vector from a basis of U is orthogonal to every vector from a basis of V:

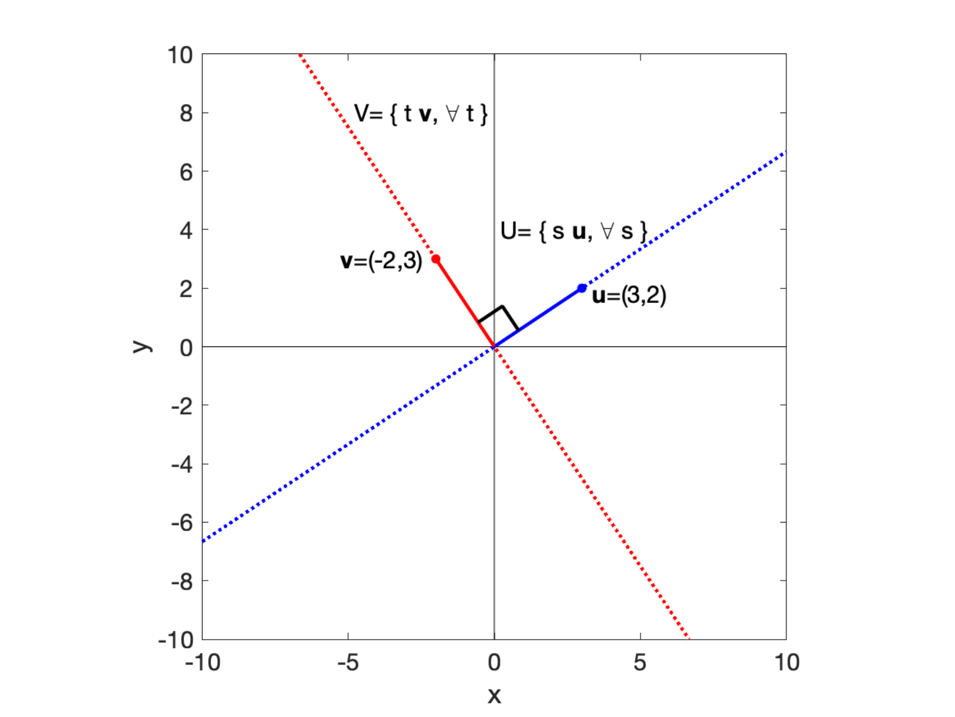
$$S = \{\vec{s}_1, \vec{s}_2, \cdots, \vec{s}_m\}, \quad U = span(S), \quad T = \{\vec{t}_1, \vec{t}_2, \cdots, \vec{t}_n\}, \quad V = span(T)$$

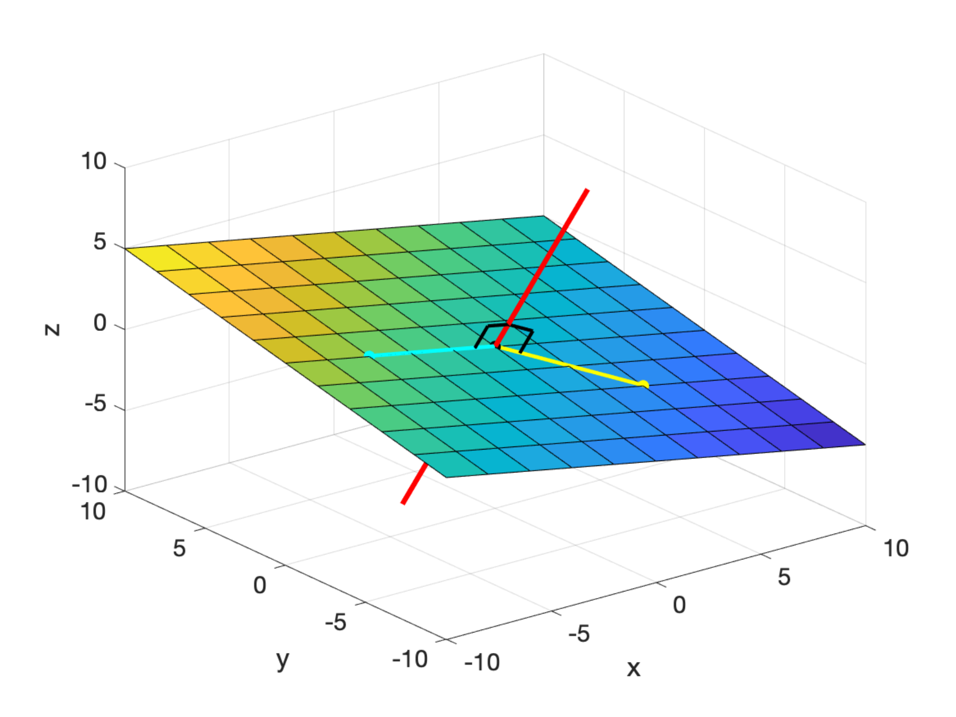
$$U \perp V \leftrightarrow \vec{s}_i^T \vec{t}_j = 0, \quad \forall \vec{s}_i \in S, \quad \forall \vec{t}_j \in T$$

$$\begin{aligned}
\vec{u}^T \vec{v} &= (a_1 \vec{s}_1^T + a_2 \vec{s}_2^T + \cdots + a_m \vec{s}_m^T)(b_1 \vec{t}_1 + b_2 \vec{t}_2 + \cdots + b_n \vec{t}_n) \\
&= a_1 b_1 \vec{s}_1^T \vec{t}_1 + a_1 b_2 \vec{s}_1^T \vec{t}_2 + \cdots + a_2 b_1 \vec{s}_2^T \vec{t}_1 + a_2 b_2 \vec{s}_2^T \vec{t}_2 + \cdots + a_m b_n \vec{s}_m^T \vec{t}_n \\
&= 0, \ \forall a_i, b_j \ \text{ iff } \ \vec{s}_i^T \vec{t}_j = 0, \forall \vec{s}_i, \vec{t}_j
\end{aligned}$$

Examples of orthogonal vector spaces.

- Two lines (through the origin) that cross at a right angle.
- A line that intersects a plane at right angles (at the origin).





The row space is orthogonal to the null space.

- Let U be the null space of a matrix, A:

$$U = \boldsymbol{N}(A) \quad \rightarrow \quad A\vec{u} = \vec{0}, \qquad \forall \vec{u} \in U$$

- Let V be the row space of A:
  - V is the column space of A^T, so $A^T\vec{y} = \vec{v}$ must have a solution:

$$V = \boldsymbol{R}(A) \rightarrow V = \boldsymbol{C}(A^T) \rightarrow \ \forall \vec{v} \in V, \qquad \exists \vec{y}: \vec{v} = A^T\vec{y}$$

- Then $\vec{v}^T\vec{u}$ must be zero: 0

$$\vec{v}^T\vec{u} = (A^T\vec{y})^T\vec{u} = (\vec{y}^T A)\vec{u} = \vec{y}^T(A\vec{u}) = \vec{y}^T\vec{0} = 0$$

$$\boldsymbol{R}(A) \perp \boldsymbol{N}(A)$$

The row space is the orthogonal complement of the null space.

- Two vector spaces U and V are **orthogonal complements** of one another **iif** they are orthogonal and together they span the parent space:

$$V = U^{\perp}, \qquad U = V^{\perp} \quad \leftrightarrow \quad U \perp V, \quad \dim(U) \oplus \dim(V) = \mathbb{R}^n, \qquad U, V \subseteq \mathbb{R}^n$$

- The row space and the null space share the **same parent space**:

$$A \in \mathbb{R}^{m \times n} \rightarrow \boldsymbol{R}(A) \subseteq \mathbb{R}^n, \boldsymbol{N}(A) \subseteq \mathbb{R}^n$$

- The row space and the null space are orthogonal:

$$\boldsymbol{R}(A) \perp \boldsymbol{N}(A)$$

- The dimensions of the row and null spaces sum to that of their parent space:

$$\dim\big(\boldsymbol{R}(A)\big) + \dim\big(\boldsymbol{N}(A)\big) = \dim(\mathbb{R}^n) \leftrightarrow \operatorname{rank}(A) + \operatorname{nullity}(A) = n$$

$$\boldsymbol{R}(A) = \boldsymbol{N}(A)^{\perp}, \qquad \boldsymbol{N}(A) = \boldsymbol{R}(A)^{\perp}$$

The column space is the orthogonal complement of the left null space.

- The column space of A is just the row space of A^T.

- The column space and the left null space share the same parent space:
$$A \in \mathbb{R}^{m \times n}. \quad \rightarrow \quad \boldsymbol{C}(A) \subseteq \mathbb{R}^m, \qquad \boldsymbol{N}(A^T) \subseteq \mathbb{R}^m$$

- The column space and the left null space are orthogonal:
$$\boldsymbol{C}(A) = \boldsymbol{R}(A^T) \perp \boldsymbol{N}(A^T)$$

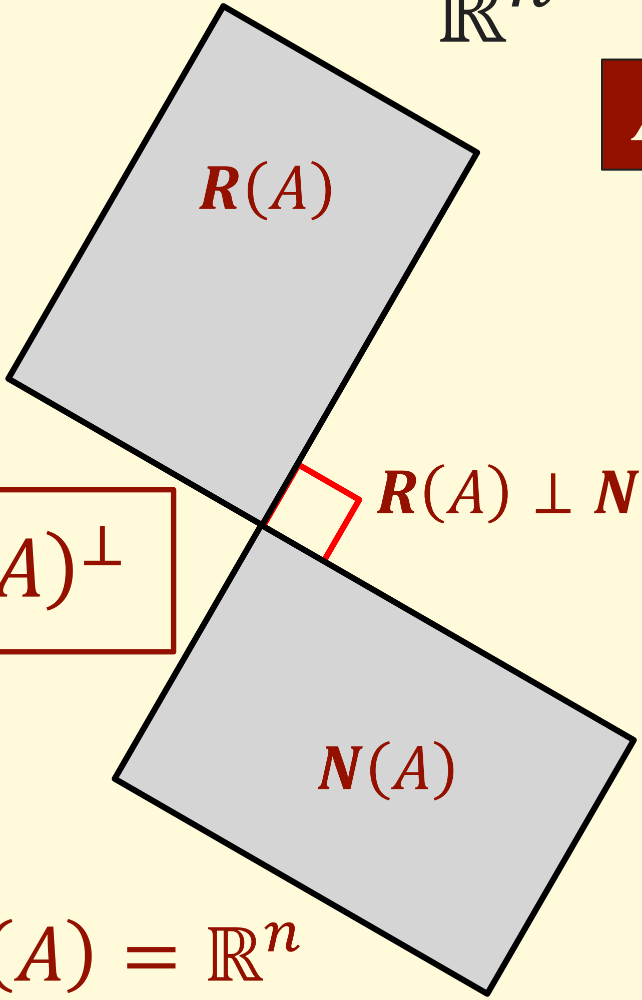
- The dimensions of the column and left null spaces sum to that of their parent space:
$$\dim\big(\boldsymbol{C}(A)\big) + \dim\big(\boldsymbol{N}(A^T)\big) = \dim(\mathbb{R}^m)$$

$$\boldsymbol{C}(A) = \boldsymbol{N}(A^T)^{\perp}, \qquad \boldsymbol{N}(A^T) = \boldsymbol{C}(A)^{\perp}$$

The fundamental matrix subspaces.

$\mathbb{R}^n$ | $\mathbb{R}^m$

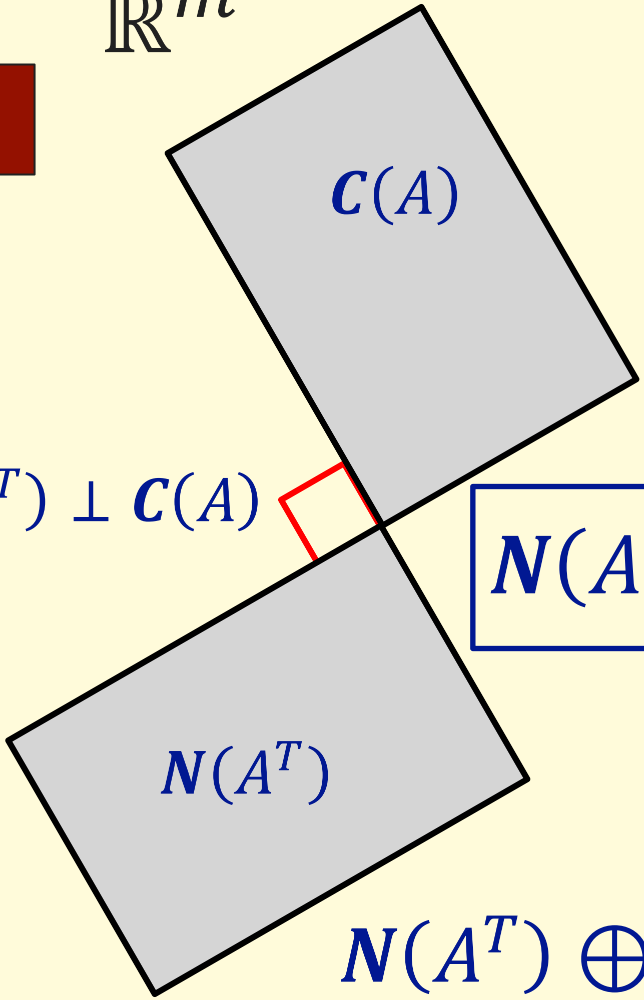
$A \in \mathbb{R}^{m \times n}$



$\boxed{\boldsymbol{R}(A) = \boldsymbol{N}(A)^{\perp}}$

$\boldsymbol{R}(A) \perp \boldsymbol{N}(A)$

$\boldsymbol{R}(A) \oplus \boldsymbol{N}(A) = \mathbb{R}^n$



$\boldsymbol{N}(A^T) \perp \boldsymbol{C}(A)$

$\boxed{\boldsymbol{N}(A^T) = \boldsymbol{C}(A)^{\perp}}$

$\boldsymbol{N}(A^T) \oplus \boldsymbol{C}(A) = \mathbb{R}^m$

Unit 8.2

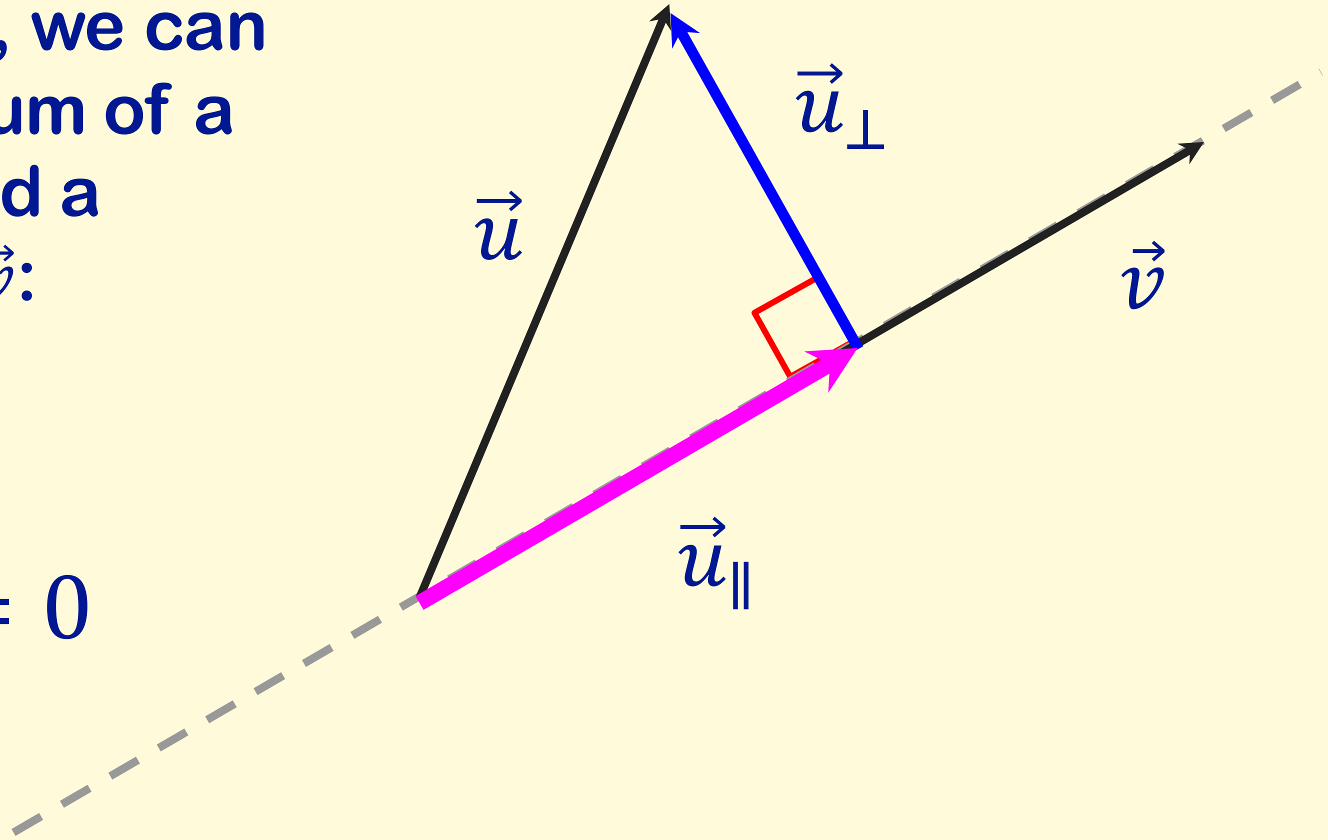
Projections and Least Squares

The projection of a vector onto a vector.

- Given a two vectors, $\vec{u}$ and $\vec{v}$, we can always write $\vec{u}$ as a unique sum of a vector that is parallel to $\vec{v}$ and a vector that is orthogonal to $\vec{v}$:

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

$$\vec{u}_{\parallel} = k\vec{v}, \qquad \vec{v}^T\vec{u}_{\perp} = 0$$



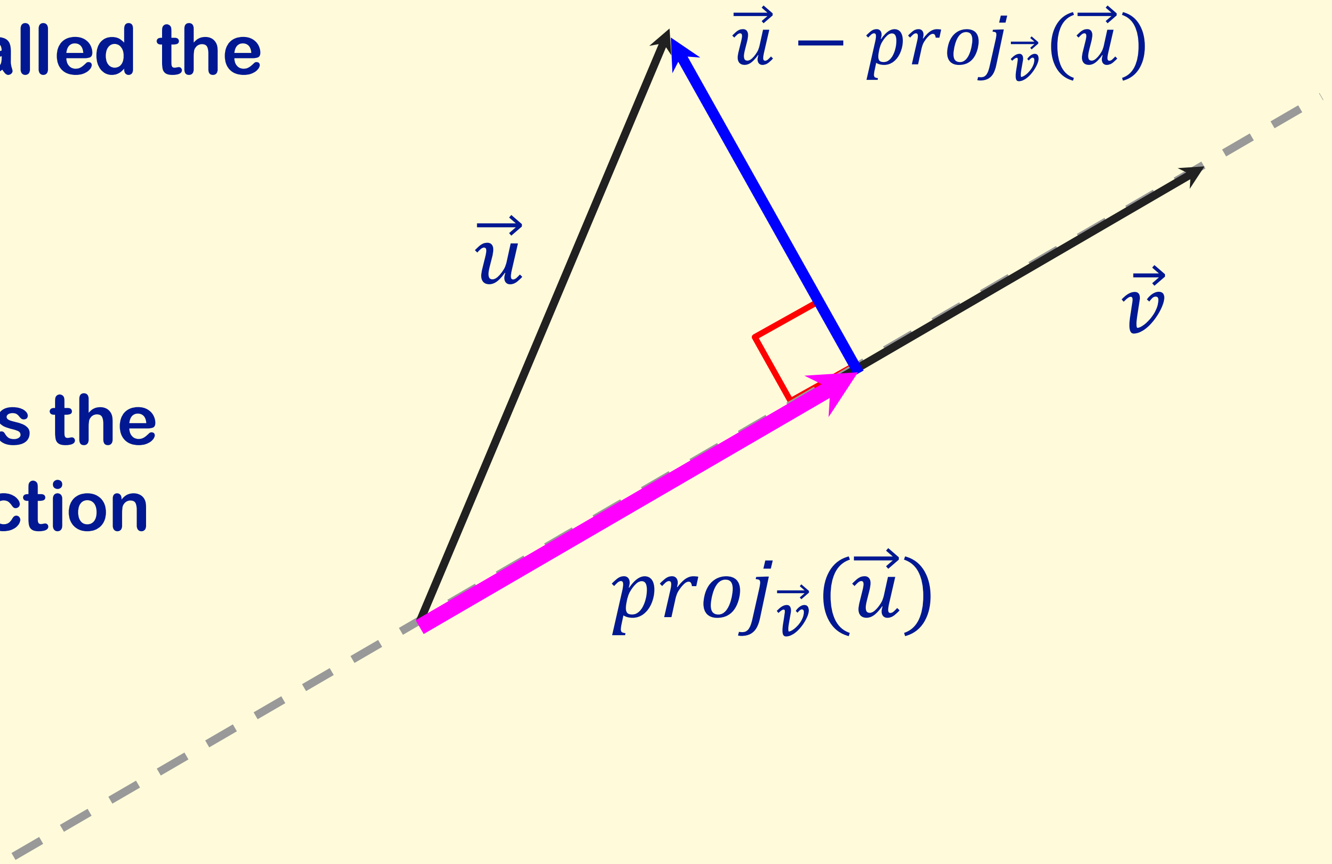
The projection of a vector onto a vector.

- The parallel component is called the **projection of $\vec{u}$ onto $\vec{v}$.**

$$\vec{u}_{\parallel} = proj_{\vec{v}}(\vec{u})$$

- The orthogonal component is the difference of $\vec{u}$ and its projection onto $\vec{v}$:

$$\vec{u}_{\perp} = \vec{u} - proj_{\vec{v}}(\vec{u})$$



The projection of a vector onto a vector.

- How do we find the projection?

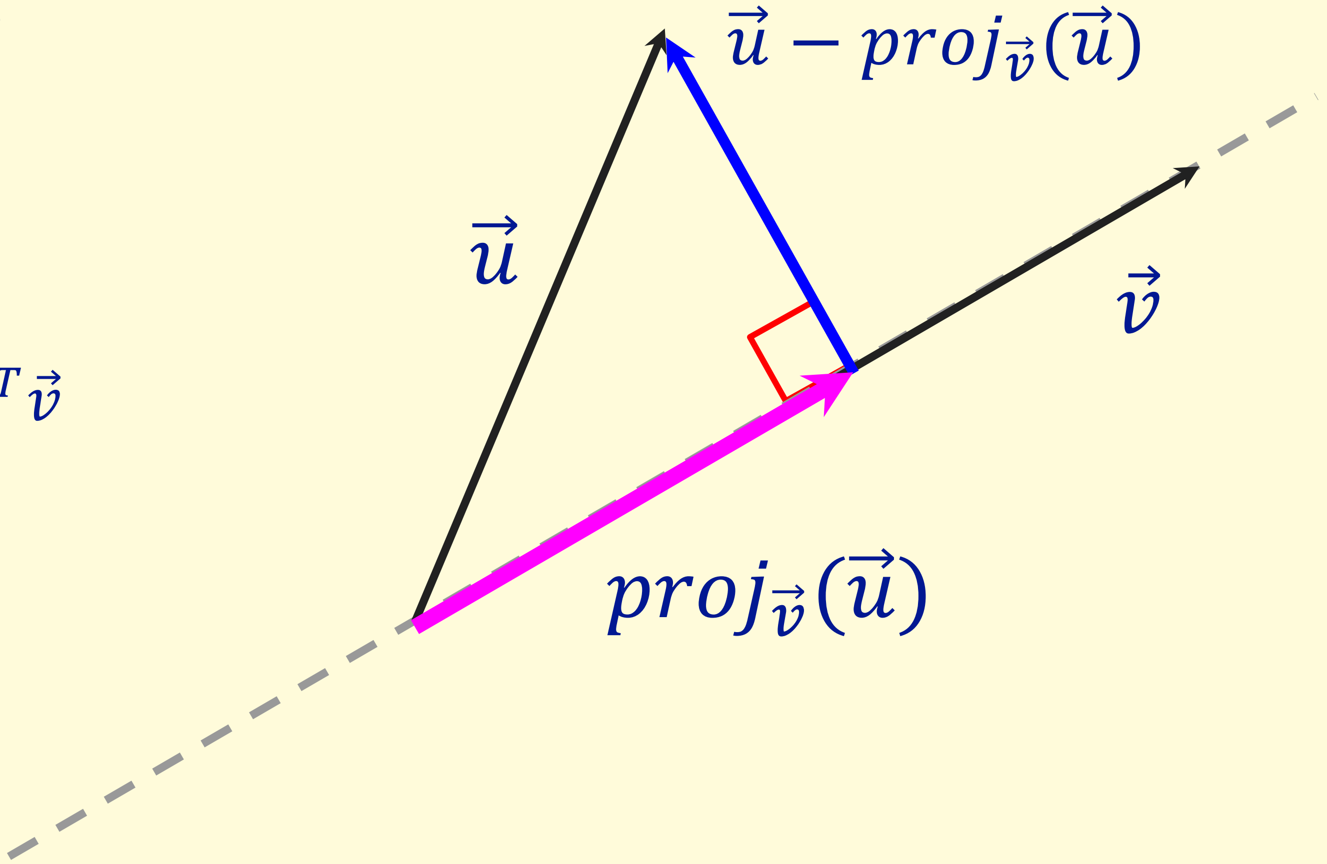
$$proj_{\vec{v}}(\vec{u}) = k\vec{v}$$

$$\vec{v}^T(\vec{u} - proj_{\vec{v}}(\vec{u})\,) = 0$$

$$\vec{v}^T(\vec{u} - k\vec{v}) = 0 \rightarrow \vec{v}^T\vec{u} = k\vec{v}^T\vec{v}$$

$$k = \frac{\vec{v}^T\vec{u}}{\vec{v}^T\vec{v}}$$

$$proj_{\vec{v}}(\vec{u}) = \left(\frac{\vec{v}^T\vec{u}}{\vec{v}^T\vec{v}}\right)\vec{v}$$

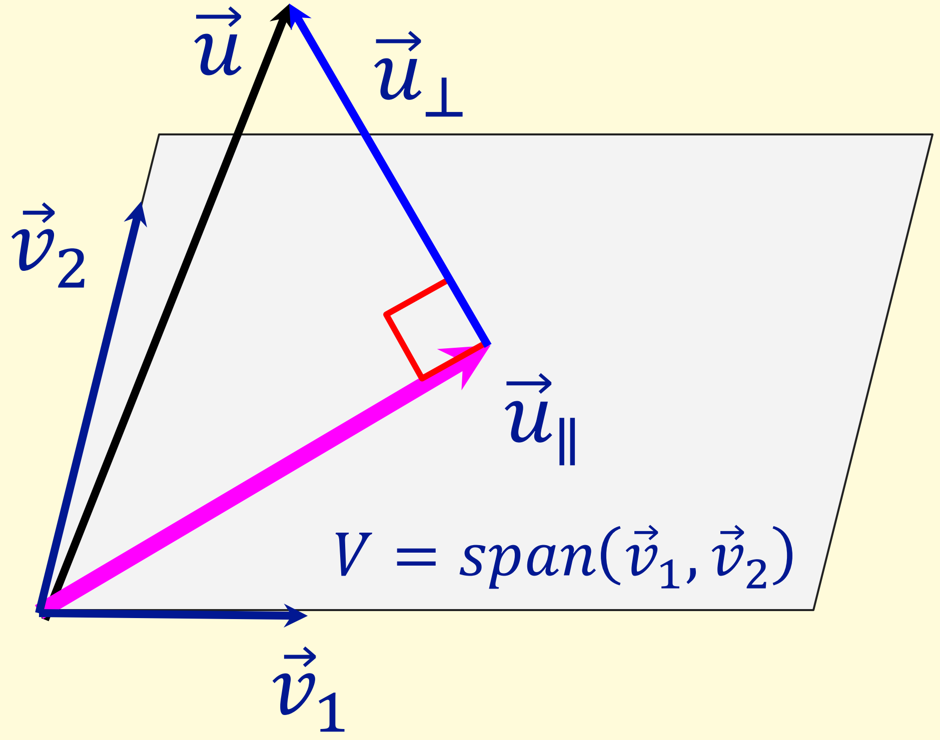


The projection of a vector onto a vector space.

- Given a vector, $\vec{u}$, and a vector space V, we can always write $\vec{u}$ as a unique sum of a vector that is in V and a vector that is orthogonal to V:

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

$$\vec{u}_{\parallel} \in V,$$
$$\vec{v}^T \vec{u}_{\perp} = 0, \forall \vec{v} \in V$$



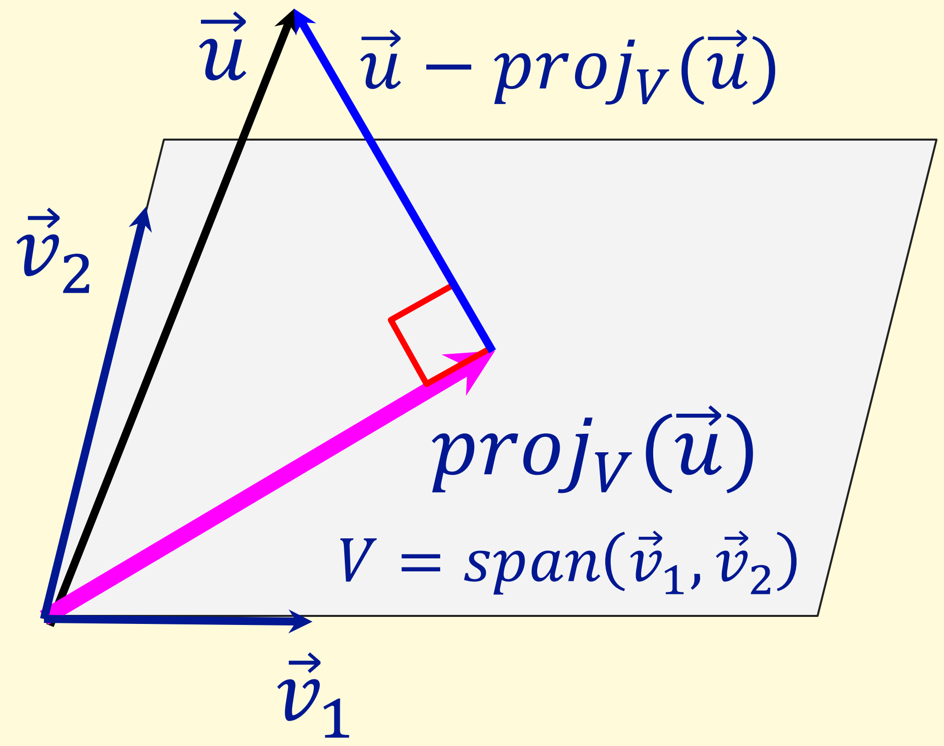
The projection of a vector onto a vector space.

- The parallel component is called the **projection of $\vec{u}$ onto V.**

$$\vec{u}_{\parallel} = proj_V(\vec{u})$$

- The orthogonal component is the difference of $\vec{u}$ and its projection onto $\vec{v}$:

$$\vec{u}_{\perp} = \vec{u} - proj_V(\vec{u})$$



The projection of a vector onto a vector space.

$$A = [\vec{v}_1 \quad \vec{v}_2]$$

$$\left.\begin{array}{l} \vec{v}_1^T(\vec{u} - proj_V(\vec{u})) = 0 \\ \vec{v}_2^T(\vec{u} - proj_V(\vec{u})) = 0 \end{array}\right. \rightarrow A^T(\vec{u} - proj_V(\vec{u})) = \vec{0}$$

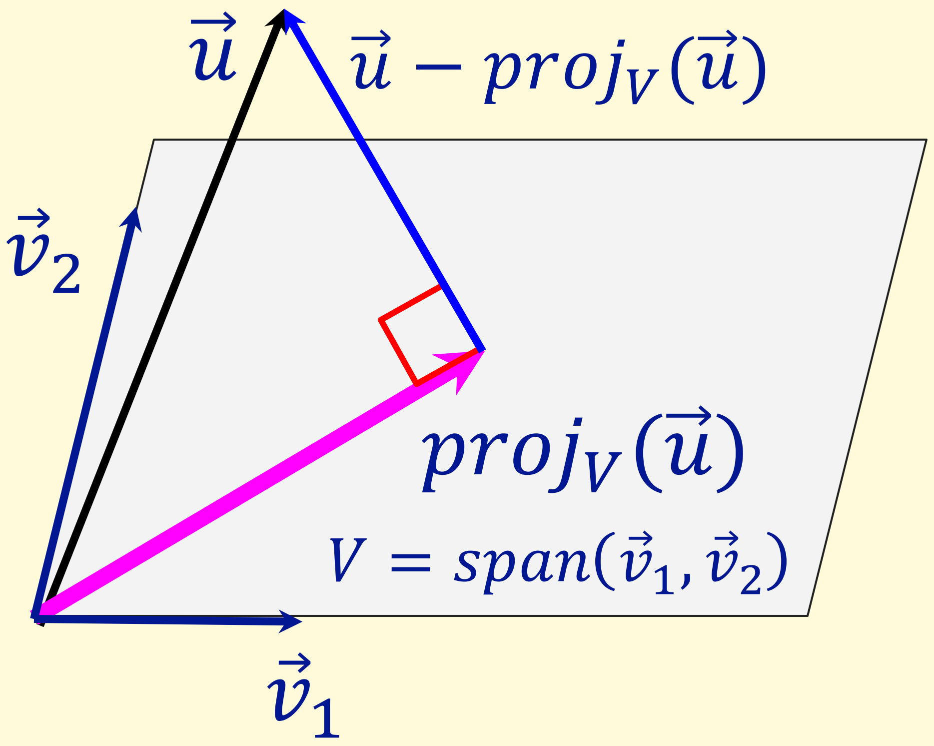
$$proj_V(\vec{u}) \in span(\vec{v}_1, \vec{v}_2) \quad \rightarrow \quad proj_V(\vec{u}) \in \boldsymbol{C}(A)$$

$$\exists \vec{w}: A\vec{w} = proj_V(\vec{u})$$

$$A^T(\vec{u} - proj_V(\vec{u})) = \vec{0} \quad \rightarrow \quad A^T\vec{u} = A^T A\vec{w}$$

$$\vec{w} = (A^T A)^{-1} A^T \vec{u}$$

$$proj_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u}$$



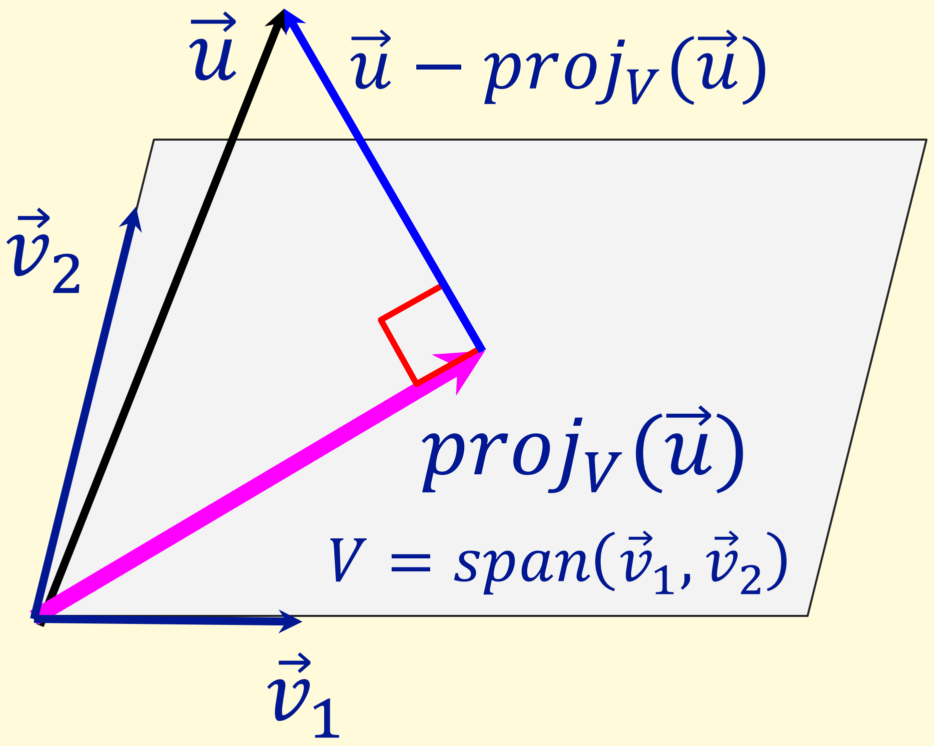
The projection of a vector onto a vector space.

- To find the projection of $\vec{u}$ onto a vector space V.
- Define a matrix A whose columns are a basis for V.
- Then:

$$proj_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u}$$

Note, $proj_{\vec{v}}(\vec{u})$ is just a special case:

$$proj_{\vec{v}}(\vec{u}) = \left(\frac{\vec{v}^T \vec{u}}{\vec{v}^T \vec{v}}\right)\vec{v} = \vec{v}(\vec{v}^T \vec{v})^{-1} \vec{v}^T \vec{u}$$

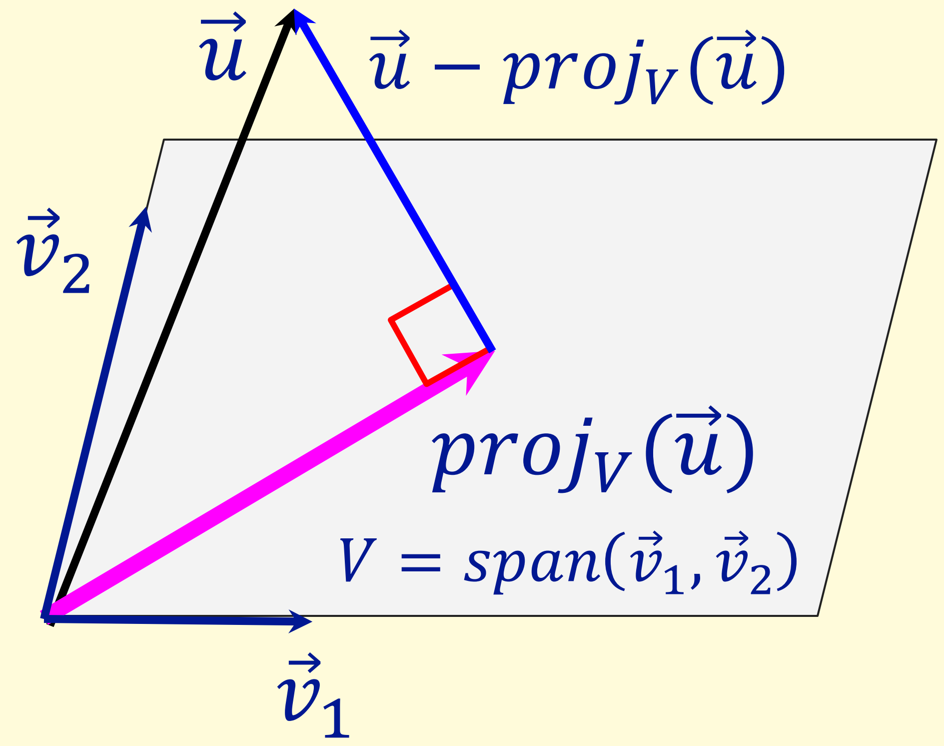


The meaning of the projection.

- The projection $proj_V(\vec{u})$ is the point in V that is closest to $\vec{u}$.

$$\min_{\vec{v}\in V} \|\vec{u}-\vec{v}\| \rightarrow proj_V(\vec{u})$$

- The orthogonal component is the deviation of $\vec{u}$ from this closest point.



An approximate solution to an overdetermined system.

- When the echelon forms of a coefficient matrix include zero rows, some right-hand sides have no solution:

$$A\vec{x} = \vec{b}, \qquad \vec{b} \notin \boldsymbol{C}(A)$$

$$\begin{bmatrix} A & \vec{b} \end{bmatrix} \rightarrow \begin{bmatrix} U & \vec{c} \end{bmatrix}, \qquad rank(\begin{bmatrix} U & \vec{c} \end{bmatrix}) > rank(U)$$

- A well-defined way to get an approximate solution is to find the solution that yields a right-hand side as close as possible to $\vec{b}$:

$$\min_{\vec{x},\vec{p}} \left\| \vec{b} - \vec{p} \right\| \;\; s.t. \;\; A\vec{x} = \vec{p}$$

An approximate solution to an overdetermined system.

- The right-hand side for the approximate solution must be in the column space of A:

$$\exists \vec{x}: A\vec{x} = \vec{p} \rightarrow \vec{p} \in \boldsymbol{C}(A)$$

- The point closest to $\vec{b}$ that is in the column space A is the projection of $\vec{b}$ onto the column space:

$$\min_{\vec{p}} \left\| \vec{b} - \vec{p} \right\| \rightarrow \vec{p} = proj_{\boldsymbol{C}(A)}\left(\vec{b}\right)$$

An approximate solution to an overdetermined system.

- Given an inconsistent system:

$$A\vec{x} = \vec{b}, \qquad \vec{b} \notin \boldsymbol{C}(A)$$

- The solution that yields a right-hand side as close as possible to $\vec{b}$ is given by solving:

$$A\vec{x} = proj_{\boldsymbol{C}(A)}(\vec{b})$$

- If the columns of A are linearly independent:

$$proj_{\boldsymbol{C}(A)}(\vec{b}) = A(A^T A)^{-1}A^T\vec{b} \rightarrow A\vec{x} = A(A^T A)^{-1}A^T\vec{b}$$

$$\vec{x} = (A^T A)^{-1}A^T\vec{b}$$

The pseudoinverse of a full column rank matrix.

- For any matrix, $A,$ the columns are linearly independent if and only if A is full column rank:

$$\mathrm{rank}(A) = n, A \in \mathbb{R}^{m \times n}$$

- Define a **pseudoinverse**:

$$A^+ = (A^T A)^{-1} A^T$$

- A^+ is a left inverse:

$$A^+ A = (A^T A)^{-1} A^T A = I$$

$$A A^+ = A(A^T A)^{-1} A^T = P \neq I$$

- For a nonsingular matrix, A^+ reduces to the inverse:

$$A^+ = A^{-1} (A^T)^{-1} A^T = A^{-1}$$

The pseudoinverse of a full column rank matrix.

- Define a pseudoinverse for full column rank matrices:

$$\operatorname{rank}(A) = n, A \in \mathbb{R}^{m\times n} \;\rightarrow\; A^+ = (A^TA)^{-1}A^T$$

- The pseudoinverse gives us the "best" answer for any right-hand side:

$$A\vec{x} = \vec{b} \quad \rightarrow \quad \vec{x} = A^+\vec{b}, \qquad \forall\vec{b}$$

- The pseudoinverse gives the exact solution to $A\vec{x} = \vec{b}$ if there is one, and the solution to $A\vec{x} = \vec{p}$ with the smallest error $\|\vec{b} - \vec{p}\|$ if there is $A\vec{x} = \vec{b}$ is inconsistent.

- The error vector is given by:

$$\vec{e} = \vec{b} - \vec{p} = \vec{b} - AA^+\vec{b} = (I - P)\vec{b}$$

Quality of the approximate solution.

- The pseudoinverse gives the best approximate answer to an inconsistent system, where best is defined as minimizing the length of $\vec{e} = \vec{b} - \vec{p}$
- The is the "least squares" solution, as what is minimized is the sum of the squared error for every element of $\vec{b}$.
- However, this "**best**" approximation is not necessarily a "**good**" approximation.
  - Error may be very large.
  - Approximate solution may violate real-world constraints.
  - The least squares approximation may not be the best for the problem.

An approximate solutions to an underdetermined inconsistent system.

- Given an inconsistent system:

$$A\vec{x} = \vec{b}, \qquad \vec{b} \notin \boldsymbol{C}(A)$$

- The solution that yields a right-hand side as close as possible to $\vec{b}$ is given by solving:

$$A\vec{x} = proj_{\boldsymbol{C}(A)}(\vec{b})$$

- What if the columns of are linearly dependent?

$$\operatorname{rank}(A) < n, \qquad A \in \mathbb{R}^{m \times n}$$

$$\operatorname{rank}(A^T A) < n, \qquad (A^T A) \in \mathbb{R}^{n \times n}$$

$$proj_{\boldsymbol{C}(A)}(\vec{b}) \neq A(A^T A)^{-1} A^T \vec{b}$$

An approximate solutions to an underdetermined inconsistent system.

- Given an inconsistent system of less than full column rank:

$$A\vec{x} = \vec{b}, \qquad \vec{b} \notin \boldsymbol{C}(A), \qquad \text{rank}(A) < n, \qquad A \in \mathbb{R}^{m\times n}$$

- Let $\hat{A}$, be the matrix derived from A by removing all free variable columns; the columns of $\hat{A}$ are a basis for $\boldsymbol{C}(A)$.

$$\boldsymbol{C}(A) = \boldsymbol{C}(\hat{A}) \rightarrow proj_{\boldsymbol{C}(A)}(\vec{b}) = proj_{\boldsymbol{C}(\hat{A})}(\vec{b})$$

$$proj_{\boldsymbol{C}(A)}(\vec{b}) = \hat{A}\left(\hat{A}^T\hat{A}\right)^{-1}\hat{A}^T\vec{b}$$

- Solve the system:

$$A\vec{x} = \hat{A}\left(\hat{A}^T\hat{A}\right)^{-1}\hat{A}^T\vec{b}$$

- As A has free variables, this will be a general approximate solution!

Polynomial Regression

- Consider a set of data, with m data points each consisting of a pair of values:

$$\{(x_1, y_1), (x_1, y_1), , (x_m, y_m)\}$$

- An n[th]-order polynomial model for the data attempts to achieve:

$$y_i = a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n, \forall i \in [1, m]$$

- A matrix representation is:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 & & x_1^n \\ 1 & x_2 & x_2^2 & & x_2^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_m & x_m^2 & & x_m^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

- In general $m \gg n$ (many rows, few columns): highly overdetermined.

Polynomial Regression

- **Given a set of data:**

$$\{(x_1, y_1\ ), (x_1, y_1), , (x_m, y_m)\}$$

- **Define:**

$$A = [\vec{c}_0 \quad \vec{c}_1 \quad \vec{c}_2 \quad \cdots \quad \vec{c}_n], \vec{c}_j = \begin{bmatrix} x_1^j \\ x_2^j \\ \vdots \\ x_m^j \end{bmatrix}, \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \vec{q} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix}$$

- **The general model is:**

$$A\vec{q} = \vec{y}$$

- **The coefficients for the minimum least-squared error model are:**

$$\vec{q} = A^+\vec{y}, \qquad A^+ = (A^T A)^{-1} A^T$$

Unit 8.2.1

Worked Examples using Projections and Least Squares

Find the projection.

- Find the projection of the vector $\vec{u}$ onto the vector space spanned by $\vec{v}_1$ and $\vec{v}_2$, where:

$$\vec{u} = (1,2,3,4), \qquad \vec{v}_1 = (1,1,1,1), \qquad \vec{v}_2 = (1,2,2,1)$$

Find the projection.

$$proj_V(\vec{u}) = A(A^TA)^{-1}A^T\vec{u}, \qquad A = [\vec{v}_1 \quad \vec{v}_2]$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \rightarrow A^TA = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 10 \end{bmatrix}$$

$$[A^TA \quad I] = \begin{bmatrix} 4 & 6 & 1 & 0 \\ 6 & 10 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 6 & 1 & 0 \\ 0 & 1 & -3/2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 4 & 0 & 10 & -6 \\ 0 & 1 & -3/2 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 5/2 & -3/2 \\ 0 & 1 & -3/2 & 1 \end{bmatrix} = [I \quad (A^TA)^{-1}]$$

Find the projection.

$$proj_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u}, \qquad A = [\vec{v}_1 \quad \vec{v}_2]$$

$$(A^T A)^{-1} = \begin{bmatrix} 5/2 & -3/2 \\ -3/2 & 1 \end{bmatrix}, \vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$A^T \vec{u} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ 15 \end{bmatrix}$$

$$(A^T A)^{-1} A^T \vec{u} = \begin{bmatrix} 5/2 & -3/2 \\ -3/2 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 15 \end{bmatrix} = \begin{bmatrix} 5/2 \\ 0 \end{bmatrix}$$

Find the projection.

$$proj_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u}, \qquad A = [\vec{v}_1 \quad \vec{v}_2]$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}, (A^T A)^{-1} A^T \vec{u} = \begin{bmatrix} 5/2 \\ 0 \end{bmatrix}$$

$$proj_V(\vec{u}) = A(A^T A)^{-1} A^T \vec{u} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 5/2 \\ 5/2 \\ 5/2 \\ 5/2 \end{bmatrix} = \left(\frac{5}{2}\right) \vec{v}_1$$

Least squares solution.

- Find the best (in a least-squares sense) approximate solution to the inconsistent system:

$$\begin{array}{llll} x_1 & -x_2 & +x_3 & = 5 \\ x_1 & & +x_3 & = 5 \\ x_1 & +x_2 & & = 5 \\ & x_2 & +x_3 & = 5 \end{array}$$

Least squares solution.

$$\begin{aligned} x_1 \quad -x_2 \quad +x_3 &= 5 \\ x_1 \qquad\quad\;\; +x_3 &= 5 \\ x_1 \quad +x_2 \qquad\quad &= 5 \\ x_2 \quad +x_3 &= 5 \end{aligned} \rightarrow A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

$$A\vec{x} = \vec{b} \rightarrow A\vec{x} = proj_{C(A)}(\vec{b}) = A(A^T A)^{-1} A^T \vec{b} \rightarrow \vec{x} = (A^T A)^{-1} A^T \vec{b}$$

$$(A^T A) = \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 3 \end{bmatrix}$$

Least squares solution.

$$\vec{x} = (A^T A)^{-1} A^T \vec{b}, A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

$$[A^T A \quad I] = \begin{bmatrix} 3 & 0 & 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 2 & 0 & 3 & 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 3 & 0 & 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & {}^{5}/_{3} & -{}^{2}/_{3} & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 3 & 0 & 2 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -{}^{2}/_{5} & 0 & {}^{3}/_{5} \end{bmatrix}$$

$$\to \begin{bmatrix} 3 & 0 & 0 & {}^{9}/_{5} & 0 & -{}^{6}/_{5} \\ 0 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -{}^{2}/_{5} & 0 & {}^{3}/_{5} \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & {}^{3}/_{5} & 0 & -{}^{2}/_{5} \\ 0 & 1 & 0 & 0 & {}^{1}/_{3} & 0 \\ 0 & 0 & 1 & -{}^{2}/_{5} & 0 & {}^{3}/_{5} \end{bmatrix} = [I \quad (A^T A)^{-1}]$$

Least squares solution.

$$\vec{x} = (A^T A)^{-1} A^T \vec{b}, A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

$$A^T \vec{b} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \\ 15 \end{bmatrix}$$

$$\vec{x} = (A^T A)^{-1} A^T \vec{b} = \begin{bmatrix} {}^{3}/_{5} & 0 & {}^{-2}/_{5} \\ 0 & {}^{1}/_{3} & 0 \\ {}^{-2}/_{5} & 0 & {}^{3}/_{5} \end{bmatrix} \begin{bmatrix} 15 \\ 5 \\ 15 \end{bmatrix} = \begin{bmatrix} 3 \\ {}^{5}/_{3} \\ 3 \end{bmatrix}$$

Least squares solution.

$$A\vec{x} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5/3 \\ 3 \end{bmatrix} = \begin{bmatrix} 13/3 \\ 6 \\ 14/3 \\ 14/3 \end{bmatrix} \approx \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

$$\vec{e} = \begin{bmatrix} 13/3 \\ 6 \\ 14/3 \\ 14/3 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 1 \\ -1/3 \\ -1/3 \end{bmatrix}, \|\vec{e}\| = \frac{\sqrt{15}}{3} = \sqrt{\frac{5}{3}}$$

Find the pseudoinverse.

- Use the pseudoinverse to solve the following system with multiple right-hand sides.
- Note whether each solution is exact or approximate.

$$A\vec{x} = \vec{b}_i, A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}, \vec{b}_1 = \begin{bmatrix} 15 \\ 15 \\ 15 \\ 15 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 20 \\ 5 \\ 30 \\ 20 \end{bmatrix}, \vec{b}_3 = \begin{bmatrix} 15 \\ 30 \\ 30 \\ 15 \end{bmatrix}$$

Find the pseudoinverse.

$$A\vec{x} = \vec{b} \rightarrow \vec{x} = A^{+}\vec{b}, A^{+} = (A^{T}A)^{-1}A^{T}$$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix} \rightarrow A^{T}A = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 9 & 6 \\ 6 & 9 \end{bmatrix}$$

$$[A^{T}A \quad I] = \begin{bmatrix} 9 & 6 & 1 & 0 \\ 6 & 9 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 9 & 6 & 1 & 0 \\ 0 & 5 & -2/3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 9 & 6 & 1 & 0 \\ 0 & 1 & -2/15 & 1/5 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 9 & 0 & 9/5 & -6/5 \\ 0 & 1 & -2/15 & 1/5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1/5 & -2/15 \\ 0 & 1 & -2/15 & 1/5 \end{bmatrix} = [I \quad (A^{T}A)^{-1}]$$

Find the pseudoinverse.

$$A\vec{x} = \vec{b} \rightarrow \vec{x} = A^{+}\vec{b}, A^{+} = (A^{T}A)^{-1}A^{T}$$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}, (A^{T}A)^{-1} = \begin{bmatrix} {}^{1}/_{5} & {}^{-2}/_{15} \\ {}^{-2}/_{15} & {}^{1}/_{5} \end{bmatrix}$$

$$A^{+} = (A^{T}A)^{-1}A^{T} = \begin{bmatrix} {}^{1}/_{5} & {}^{-2}/_{15} \\ {}^{-2}/_{15} & {}^{1}/_{5} \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} {}^{-1}/_{15} & {}^{-2}/_{15} & {}^{2}/_{15} & {}^{2}/_{5} \\ {}^{4}/_{15} & {}^{1}/_{5} & {}^{2}/_{15} & {}^{-4}/_{15} \end{bmatrix}$$

$$A^{+} = \left(\frac{1}{15}\right) \begin{bmatrix} -1 & -2 & 2 & 6 \\ 4 & 3 & 2 & -4 \end{bmatrix}$$

Find the pseudoinverse.

$$\vec{x}_i = A^+\vec{b}_i, \vec{b}_1 = \begin{bmatrix} 15 \\ 15 \\ 15 \\ 15 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 20 \\ 5 \\ 30 \\ 20 \end{bmatrix}, \vec{b}_3 = \begin{bmatrix} 15 \\ 30 \\ 30 \\ 15 \end{bmatrix}$$

$$\vec{x}_1 = A^+\vec{b}_1 = \left(\frac{1}{15}\right)\begin{bmatrix} -1 & -2 & 2 & 6 \\ 4 & 3 & 2 & -4 \end{bmatrix}\begin{bmatrix} 15 \\ 15 \\ 15 \\ 15 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

$$A\vec{x}_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \\ 20 \\ 10 \end{bmatrix}, \vec{e}_1 = A\vec{x}_1 - \vec{b}_1 = \begin{bmatrix} 0 \\ -10 \\ +5 \\ -5 \end{bmatrix}$$

Find the pseudoinverse.

$$\vec{x}_i = A^+\vec{b}_i, \vec{b}_1 = \begin{bmatrix} 15 \\ 15 \\ 15 \\ 15 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 20 \\ 5 \\ 30 \\ 20 \end{bmatrix}, \vec{b}_3 = \begin{bmatrix} 15 \\ 30 \\ 30 \\ 15 \end{bmatrix}$$

$$\vec{x}_2 = A^+\vec{b}_2 = \left(\frac{1}{15}\right)\begin{bmatrix} -1 & -2 & 2 & 6 \\ 4 & 3 & 2 & -4 \end{bmatrix}\begin{bmatrix} 20 \\ 5 \\ 30 \\ 20 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

$$A\vec{x}_2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 20 \\ 5 \\ 30 \\ 20 \end{bmatrix}, \vec{e}_2 = A\vec{x}_2 - \vec{b}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Find the pseudoinverse.

$$\vec{x}_i = A^+\vec{b}_i, \vec{b}_1 = \begin{bmatrix} 15 \\ 15 \\ 15 \\ 15 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 20 \\ 5 \\ 30 \\ 20 \end{bmatrix}, \vec{b}_3 = \begin{bmatrix} 15 \\ 30 \\ 30 \\ 15 \end{bmatrix}$$

$$\vec{x}_3 = A^+\vec{b}_3 = \left(\frac{1}{15}\right)\begin{bmatrix} -1 & -2 & 2 & 6 \\ 4 & 3 & 2 & -4 \end{bmatrix}\begin{bmatrix} 15 \\ 30 \\ 30 \\ 15 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$A\vec{x}_3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 5 \\ 10 \end{bmatrix} = \begin{bmatrix} 25 \\ 10 \\ 30 \\ 10 \end{bmatrix}, \vec{e}_3 = A\vec{x}_3 - \vec{b}_3 = \begin{bmatrix} +10 \\ -20 \\ 0 \\ -5 \end{bmatrix}$$

Unit 8.2.2

Solving Real World Problems with Projections and Least Squares

Office workload.

A municipality has two offices that handle construction permits:

- Each week the main office processes 600 permit applications, inspects 300 on-going projects and handles 120 complaints; the branch office processes 400 permit applications, inspects 100 on-going projects and handles 80 complaints.
- If the city has a total monthly demand of 2000 permit applications, 1000 inspections and 800 complaints, how many days must each office work to best satisfy this demand?
- Is this a good solution?
- How would this change if the complaint load was reduced by 50%?

Office workload.

A municipality has two offices that handle construction permits:

- Each week, the main office processes 600 permit applications, inspects 300 on-going projects and handles 120 complaints; the branch office processes 400 permit applications, inspects 100 on-going projects and handles 80 complaints.
- If the city has a total monthly demand of 2000 permit applications, 1000 inspections and 800 complaints, how many days must each office work to best satisfy this demand.

- Let x_1 and x_2 be the weeks worked by the main and branch offices, respectively
- Write one equation for permits, one for inspections and one for complaints:

$$\begin{aligned} 600x_1 &\quad +400x_2 &\quad = 2000 \\ 300x_1 &\quad +100x_2 &\quad = 1000 \\ 120x_1 &\quad +80x_2 &\quad = 800 \end{aligned}$$

- Simplify each equation:

$$\begin{aligned} 3x_1 &\quad +2x_2 &\quad = 10 \\ \rightarrow 3x_1 &\quad +1x_2 &\quad = 10 \\ 3x_1 &\quad +2x_2 &\quad = 20 \end{aligned}$$

Office workload.

- **Express in matrix form:**

$$\begin{matrix} 3x_1 & +2x_2 & = 10 \\ 3x_1 & +1x_2 & = 10 \\ 3x_1 & +2x_2 & = 20 \end{matrix} \rightarrow A\vec{x} = \vec{b}, A = \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix}, \vec{b} = \begin{bmatrix} 10 \\ 10 \\ 20 \end{bmatrix}$$

- **Solve with the pseudoinverse:**

$$\vec{x} = A^{+}\vec{b}, A^{+} = (A^T A)^{-1} A^T$$

Office workload.

$$A^T A = \begin{bmatrix} 3 & 3 & 3 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 27 & 15 \\ 15 & 9 \end{bmatrix}$$

$$[A^T A \quad I] = \begin{bmatrix} 27 & 15 & 1 & 0 \\ 15 & 9 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 27 & 15 & 1 & 0 \\ 0 & {}^{6}/_{9} & -{}^{5}/_{9} & 1 \end{bmatrix} \to \begin{bmatrix} 27 & 15 & 1 & 0 \\ 0 & 1 & -{}^{5}/_{6} & {}^{3}/_{2} \end{bmatrix}$$

$$\to \begin{bmatrix} 27 & 0 & {}^{27}/_{2} & -{}^{45}/_{2} \\ 0 & 1 & -{}^{5}/_{6} & {}^{3}/_{2} \end{bmatrix} \to \begin{bmatrix} 1 & 0 & {}^{1}/_{2} & -{}^{5}/_{6} \\ 0 & 1 & -{}^{5}/_{6} & {}^{3}/_{2} \end{bmatrix} = [I \quad (A^T A)^{-1}]$$

Office workload.

$$A^{+} = (A^T A)^{-1} A^T = \left(\frac{1}{6}\right)\begin{bmatrix} 3 & -5 \\ -5 & 9 \end{bmatrix}\begin{bmatrix} 3 & 3 & 3 \\ 2 & 1 & 2 \end{bmatrix} = \left(\frac{1}{6}\right)\begin{bmatrix} -1 & 4 & -1 \\ 3 & -6 & 3 \end{bmatrix}$$

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix}, \vec{b} = \begin{bmatrix} 10 \\ 10 \\ 20 \end{bmatrix}$$

$$\vec{x} = A^{+}\vec{b}$$

$$\left(\frac{1}{6}\right)\begin{bmatrix} -1 & 4 & -1 \\ 3 & -6 & 3 \end{bmatrix}\begin{bmatrix} 10 \\ 10 \\ 20 \end{bmatrix} = \left(\frac{1}{6}\right)\begin{bmatrix} 10 \\ 30 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 5 \end{bmatrix}$$

Office workload.

A municipality has two offices that handle construction permits:

- If the city has a total monthly demand of 2000 permit applications, 1000 inspections and 800 complaints, how many days must each office work to best satisfy this demand.
- Is this a good solution?
- How would this change if the complaint load was reduced by 50%?

$$\vec{x} = A^{+}\vec{b} = \begin{bmatrix} 5/3 \\ 5 \end{bmatrix}$$

- The main office needs to work 1.67 weeks (8-12 days) and the branch office needs to work 5 weeks (25-35 days)!

$$A\vec{x} = \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 5/3 \\ 5 \end{bmatrix} = \begin{bmatrix} 15 \\ 10 \\ 15 \end{bmatrix} \rightarrow \begin{bmatrix} 3000 \\ 1000 \\ 600 \end{bmatrix}$$

$$\vec{e} = \begin{bmatrix} 3000 \\ 1000 \\ 600 \end{bmatrix} - \begin{bmatrix} 2000 \\ 1000 \\ 800 \end{bmatrix} = \begin{bmatrix} 1000 \\ 0 \\ -200 \end{bmatrix} \rightarrow \begin{bmatrix} +50\% \\ 0 \\ -25\% \end{bmatrix}$$

- This isn't a great solution!

Office workload.

A municipality has two offices that handle construction permits:

- If the city has a total monthly demand of 2000 permit applications, 1000 inspections and 800 complaints, how many days must each office work to best satisfy this demand.
- Is this a good solution?
- How would this change if the complaint load was reduced by 50%?

$$\vec{b} = \begin{bmatrix} 10 \\ 10 \\ 20 \end{bmatrix} \rightarrow \vec{b} = \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix}$$

$$\vec{x} = A^{+}\vec{b}$$

$$\left(\frac{1}{6}\right)\begin{bmatrix} -1 & 4 & -1 \\ 3 & -6 & 3 \end{bmatrix}\begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix} = \left(\frac{1}{6}\right)\begin{bmatrix} 20 \\ 0 \end{bmatrix} = \begin{bmatrix} {}^{10}\!/_{3} \\ 0 \end{bmatrix}$$

$$A\vec{x} = \begin{bmatrix} 3 & 2 \\ 3 & 1 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} {}^{10}\!/_{3} \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix} = \vec{b}$$

- This gives an exact solution!
- The main office needs to work 3.33 weeks (17-24 days) and the branch office needs to work none.

Regression

- Set up, but do not solve, the following regression problems using the given data set for earnings between generations; data is average annual income for parents and their children, in thousands of dollars (and the numbers are completely made up):

Parents	$15K	$30K	$45K	$60K	$80K	$100K	$150K	$200K	$250K
Children	$16K	$29K	$43K	$55K	$82K	$130K	$190K	$280K	$400K

- For each case, give an expression both for the parameters of the fit, and for the total squared error; consider the parents' income to be the independent variable.
  a) Find the best linear fit, $y = mx + b$.
  b) Find the best linear fit through through the origin.
  c) Find the best quadratic fit, $y = ax^2 + bx + c$.
  d) Find the best quadratic fit through the origin.

Regression

- **Given a set of data:**

$$\{(x_1, y_1), (x_2, y_2), , (x_m, y_m)\}$$

- **Target model:**

$$y_i = a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n, \forall i \in [1, m]$$

$$A\vec{q} = \vec{y}, A = [\vec{c}_0 \quad \vec{c}_1 \quad \vec{c}_2 \quad \cdots \quad \vec{c}_n], \vec{c}_j = \begin{bmatrix} x_1^j \\ x_2^j \\ \vdots \\ x_m^j \end{bmatrix}, \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \vec{q} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix}$$

$$\vec{q} = A^+\vec{y}, A^+ = (A^T A)^{-1} A^T$$

Best linear fit: y = mx + b

Target model: $y = a_0 + a_1 x, a_0 = b, a_1 = m$

Data:

Parents	\$15K	\$30K	\$45K	\$60K	\$80K	\$100K	\$150K	\$200K	\$250K
Children	\$16K	\$29K	\$43K	\$55K	\$82K	\$130K	\$190K	\$280K	\$400K

$$A\vec{q} = \vec{y}, A = \begin{bmatrix} 1 & 15 \\ 1 & 30 \\ 1 & 45 \\ 1 & 60 \\ 1 & 80 \\ 1 & 100 \\ 1 & 150 \\ 1 & 200 \\ 1 & 250 \end{bmatrix}, \vec{y} = \begin{bmatrix} 16 \\ 29 \\ 43 \\ 55 \\ 82 \\ 130 \\ 190 \\ 280 \\ 400 \end{bmatrix}, \vec{q} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} b \\ m \end{bmatrix} = A^+\vec{y},\ A^+ = (A^T A)^{-1} A^T$$

Error: $\|AA^+\vec{y} - \vec{y}\|$

Best linear fit through origin, b=0

Target model: $y = a_1 x, a_1 = m$

Data:

Parents	\$15K	\$30K	\$45K	\$60K	\$80K	\$100K	\$150K	\$200K	\$250K
Children	\$16K	\$29K	\$43K	\$55K	\$82K	\$130K	\$190K	\$280K	\$400K

$$A\vec{q} = \vec{y}, A = \begin{bmatrix} 15 \\ 30 \\ 45 \\ 60 \\ 80 \\ 100 \\ 150 \\ 200 \\ 250 \end{bmatrix}, \vec{y} = \begin{bmatrix} 16 \\ 29 \\ 43 \\ 55 \\ 82 \\ 130 \\ 190 \\ 280 \\ 400 \end{bmatrix}, \vec{q} = [a_1] = [m] = A^+\vec{y},\ A^+ = (A^T A)^{-1} A^T$$

Error: $\|AA^+\vec{y} - \vec{y}\|$

Best quadratic fit: y = ax^2 + bx + c

Target model: $y = a_0 + a_1 x + a_2 x^2, a_0 = c, a_1 = b, a_2 = a$

Data:

Parents	\$15K	\$30K	\$45K	\$60K	\$80K	\$100K	\$150K	\$200K	\$250K
Children	\$16K	\$29K	\$43K	\$55K	\$82K	\$130K	\$190K	\$280K	\$400K

$$A\vec{q} = \vec{y}, A = \begin{bmatrix} 1 & 15 & 15^2 \\ 1 & 30 & 30^2 \\ 1 & 45 & 45^2 \\ 1 & 60 & 60^2 \\ 1 & 80 & 80^2 \\ 1 & 100 & 100^2 \\ 1 & 150 & 150^2 \\ 1 & 200 & 200^2 \\ 1 & 250 & 250^2 \end{bmatrix}, \vec{y} = \begin{bmatrix} 16 \\ 29 \\ 43 \\ 55 \\ 82 \\ 130 \\ 190 \\ 280 \\ 400 \end{bmatrix}, \vec{q} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} c \\ b \\ a \end{bmatrix} = A^+\vec{y},\ A^+ = (A^T A)^{-1} A^T$$

Error: $\|AA^+\vec{y} - \vec{y}\|$

Best quadratic fit through origin, c=0

Target model: $y = a_1 x + a_2 x^2, a_1 = b, a_2 = a$

Data:

Parents	\$15K	\$30K	\$45K	\$60K	\$80K	\$100K	\$150K	\$200K	\$250K
Children	\$16K	\$29K	\$43K	\$55K	\$82K	\$130K	\$190K	\$280K	\$400K

$$A\vec{q} = \vec{y}, A = \begin{bmatrix} 15 & 15^2 \\ 30 & 30^2 \\ 45 & 45^2 \\ 60 & 60^2 \\ 80 & 80^2 \\ 100 & 100^2 \\ 150 & 150^2 \\ 200 & 200^2 \\ 250 & 250^2 \end{bmatrix}, \vec{y} = \begin{bmatrix} 16 \\ 29 \\ 43 \\ 55 \\ 82 \\ 130 \\ 190 \\ 280 \\ 400 \end{bmatrix}, \vec{q} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} b \\ a \end{bmatrix} = A^+\vec{y},\ A^+ = (A^T A)^{-1} A^T$$

Error: $\|AA^+\vec{y} - \vec{y}\|$